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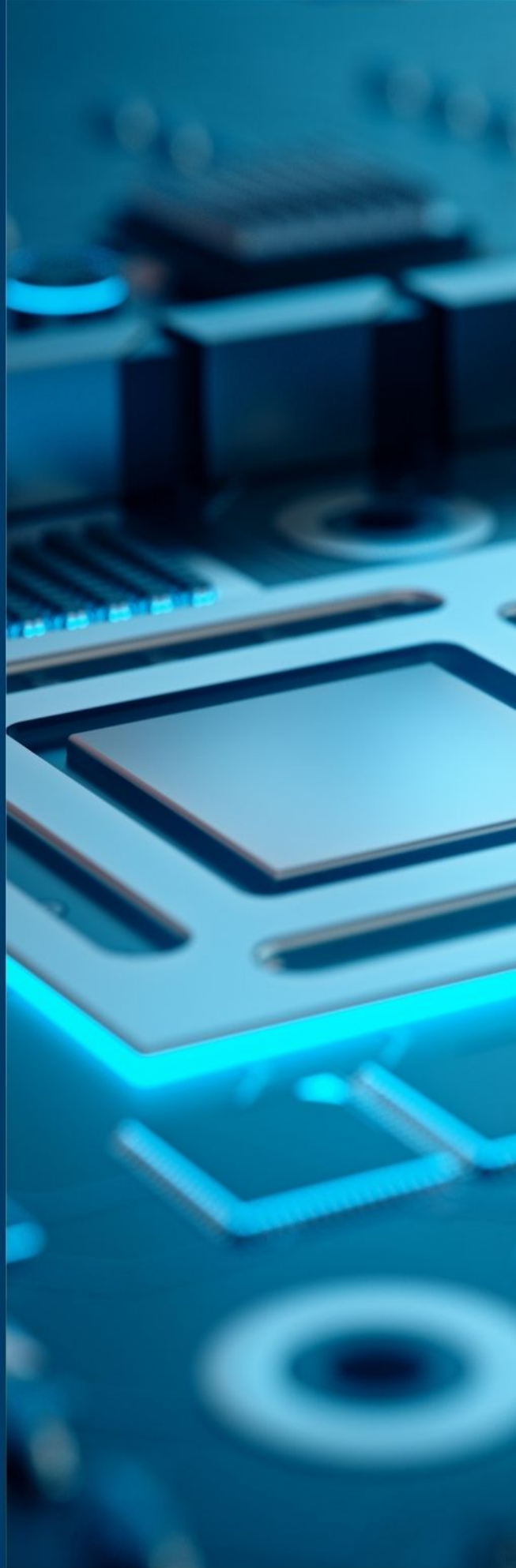
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Application of shunt active power filter for harmonic reduction and reactive power compensation in three-phase four-wire systems

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Abstract: This study deals with the analysis and implementation of compensation algorithms applied to a shunt active power filter, which uses three single-phase full-bridge converters sharing the same dc-bus voltage. The shunt filter is applied to three-phase four-wire systems, performing harmonic current suppression, reactive power compensation and power factor improvement. In addition, load unbalances compensation is also carried out. Two different control strategies are presented. In the first strategy, which is called independent current control, the currents of the three-phase power source are independently compensated performing harmonic suppression and load reactive power compensation, that is, the three-phase four-wire system is treated as three independent single-phase systems. In the second strategy, in addition to harmonic suppression and load reactive power compensation, the shunt filter also performs load unbalance compensation, resulting in sinusoidal and balanced source currents. The compensating algorithms are evaluated by means of several experimental test conditions, in order to validate the theoretical development and analyse the performance of the shunt filter.

1 Introduction

Harmonics in power supply systems have risen significantly in recent years primarily because of the increase of non-linear loads connected to the utility grid through residential, commercial and industrial customers. The power supply voltages become distorted on account of the high level of undesired harmonic currents drawn from the utility grid. In addition, drops in the fundamental voltage are caused by the interaction between the reactive current and the network impedances. To solve or minimise some power quality problems, series, shunt, and hybrid active power filters (APFs), unified power quality conditioners (UPQC) and other alternatives have been developed [1–29]. Although the series APFs are normally used to compensate utility disturbances, such as voltage sags, swells, unbalances and harmonics [22–25], shunt active power filters (SAPFs) have commonly been used for reducing harmonic currents, compensating reactive power of the load and improving power factor [1–20].

In addition to that, other problems must be considered, such as unbalanced source currents and/or excessive currents flowing through the neutral conductor, which are caused by unbalanced loads connected in three-phase four-wire systems. In other words, negative and zero sequence components exist in the three-phase source currents, helping degrade the performance of some equipment, such as transformers, for instance. For non-linear loads, therefore, the source currents contain

unbalanced fundamental and harmonic components. In addition, even with perfectly balanced single-phase non-linear loads, the third harmonic component and its multiples flow through the neutral wire. Moreover, an excessive zero sequence current can help cause damage in the neutral conductor [30].

In this work, SAPF applied to three-phase four-wire systems is performed by using three full-bridge (3F-B) single-phase converters [3, 4, 9, 18–20]. In this case, it is possible to consider each one of the three full-bridge converters as three independent single-phase SAPF sharing the same dc-bus voltage. Thereby, the degree of controllability is increased ensuring more accuracy in the current unbalance compensation.

In [20], the current synchronous detection method is applied to an SAPF, in order to obtain balanced source currents for unbalanced three-phase loads under unbalanced three-phase voltages. Although both reactive power compensating and harmonic current suppression are carried out, balanced three-phase source currents are not achieved. In addition, the proposed method is very dependent on the exact measurements of the phase voltage amplitudes. In [3], controllers based on *pq*-theory are used to control each phase dependently or independently in a three-phase four-wire system, by using also the 3F-B topology applied to an SAPF. On the other hand, compensation algorithms based on *pq*-theory can affect the performance of APFs when utility voltage disturbances, such as harmonics and unbalances occur [1, 19]. Thereby, as an alternative to

minimise or overcome the problems related to some voltage disturbances in APFs, synchronous reference frame (SRF)-based controllers have been employed [29, 31, 32].

SRF-based controllers can be used in both single- and three-phase systems. For single-phase systems they can be used by creating fictitious three-phase systems. Thereby, in three-phase four-wire systems, for instance, it is possible to create three fictitious three-phase systems, allowing the control of the three phases independently.

The main contribution of this paper is the proposition of alternative and flexible algorithms using SRF-based controllers, for compensation of sequence components (negative, zero or both) related to the utility currents. In other words, the flexibility for choosing the SRF-based controller strategy determines if the compensated three-phase source currents become balanced or unbalanced. For such purpose, two SRF-based algorithms are implemented to evaluate the behaviour and performance of the SAPF under balanced and unbalanced load conditions. Mathematical analyses of the SRF-based algorithms are presented and experimentally tested in order to confirm the theoretical development of the proposed algorithms. Furthermore, the performance of the SAPF is largely evaluated.

This paper is organised as follows: In Section 2 the SAPF topology is presented. The SRF-based controllers are discussed in Section 3 considering balanced and unbalanced load conditions. Section 4 presents the phase-locked loop (PLL) algorithm used for both synchronising the SAPF

with respect to the utility and generation of the unit vector coordinates ($\sin \theta$ and $\cos \theta$) used in the SRF-based algorithms. The performance of the SAPF is evaluated by means of experimental tests in Section 5. Finally, the conclusions are presented in Section 6.

2 Shunt APF topology

The shunt APF topology employed in this work is shown in Fig. 1. It is implemented by using three single-phase full-bridge (3F-B) VSI inverters [3, 4, 18–20]. As a consequence of using the 3F-B topology, independent phase-current compensation of the three phases is allowed and, in addition, the dc-bus voltage is reduced by a factor of $\sqrt{3}$ when compared with the four-leg topology [6]. Thus, because of the dc-bus voltage decreasing, the 3F-B topology requires reduced semiconductor voltage ratings, and, therefore, higher power rate APF applications could be feasible. As shown in Fig. 1, the 3F-B converters are sharing the same dc-bus voltage. In addition, the PWM converters are connected to the power supply system by means of three coupling inductors acting as low-pass filters (LPFs).

3 SRF-based controllers

Several methods and theories have been presented in the literature for obtaining the APF compensation current

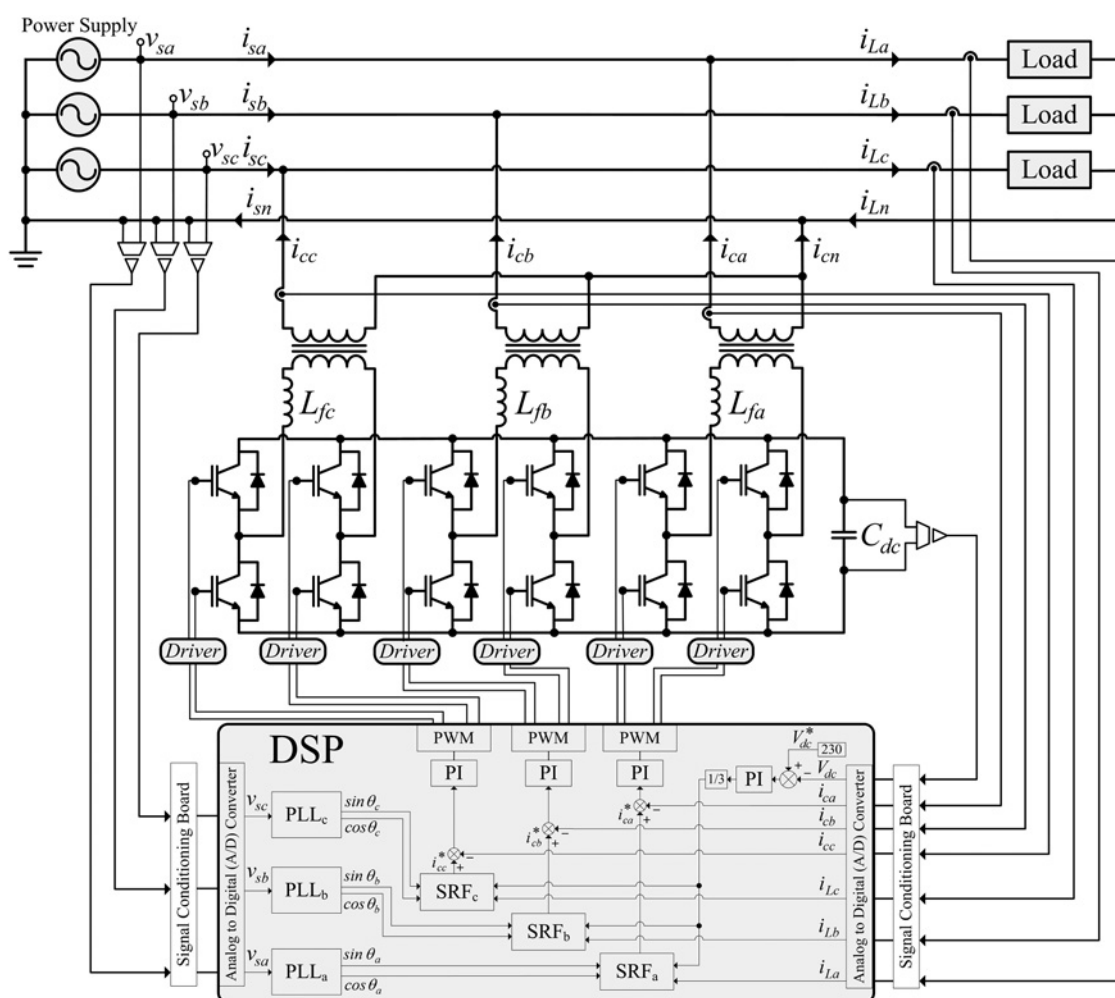


Fig. 1 Three-phase four-wire SAPF with 3F-B inverters

references [1, 3, 29, 32]. In [29], for instance, SRF-based controller was employed in a hybrid APF solution for improving passive filter performance in high power applications, whereas in [31, 32] the SRF-based controller was used to generate the sinusoidal compensating current references applied to a three-phase line-interactive UPS system. In this work, for this purpose, an algorithm based on SRF is also used.

In the SRF-based algorithm the fundamental terms of voltage and/or current of the abc -phase stationary reference frame are transformed into continuous quantities into the dq synchronous axes, in which they rotate at a synchronous speed in relation to the spatial vectors of voltage and/or current. In the dq -axes, the harmonic contents of voltage and/or current can be represented by alternate quantities, which are superposed on the continuous components. Therefore the fundamental component can be easily obtained by means of LPFs. The estimation of the utility grid phase-angle (θ) can be performed by using PLL algorithms [33–38], allowing the generation of the unit vector coordinates ($\sin \theta$ and $\cos \theta$) used in the SRF-based algorithm.

A well-known SRF-based controller used to generate the three-phase compensation current references (i_{ca}^* , i_{cb}^* , i_{cc}^*) in SAPF applications is shown in Fig. 2a. Once, the SRF control method is based on balanced three-phase loads, load unbalance compensation (LUnC) is performed, resulting in balanced and sinusoidal compensated source currents. In the dq coordinates of Fig. 2a, an LPF is used to extract the direct current id_{dc} , which represents the fundamental active component of the load currents (i_{La} , i_{Lb} , i_{Lc}). The orthogonal current iq represents the sum of both the fundamental reactive component and part of the harmonic components of the load currents. It does not need to be calculated because it is not used in the compensation algorithms. Thus, only the direct current component id is used to extract the fundamental active load currents that are used to generate the compensating reference currents of the SAPF.

The sinusoidal current references are obtained from the matrix given by (1), which provides the linear transformation from a two-phase system to a three-phase stationary reference frame system. Thereby, the compensating current references ($i_{ca,b,c}^*$) are obtained subtracting the sinusoidal current references ($i_{sa,b,c}^*$) from the respective measured load currents ($i_{La,b,c}$), as given by (2)

$$\begin{bmatrix} i_{sa}^* \\ i_{sb}^* \\ i_{sc}^* \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_{\alpha_f} \\ i_{\beta_f} \end{bmatrix} \quad (1)$$

$$[i_{ca,b,c}^*] = [i_{La,b,c}] - [i_{sa,b,c}^*] \quad (2)$$

3.1 SRF-based algorithm applied to single-phase systems

As previously mentioned, the three-phase four-wire system is treated as three single-phase systems in order to allow the controlling of each phase-current independently. Thereby, since the SRF-based controller is used to generate the current references for the SAPF compensation, three fictitious three-phase systems must be created. Thus, the new fictitious three-phase systems can also be represented by the fictitious two-phase stationary reference frame ($\alpha\beta$ -axes).

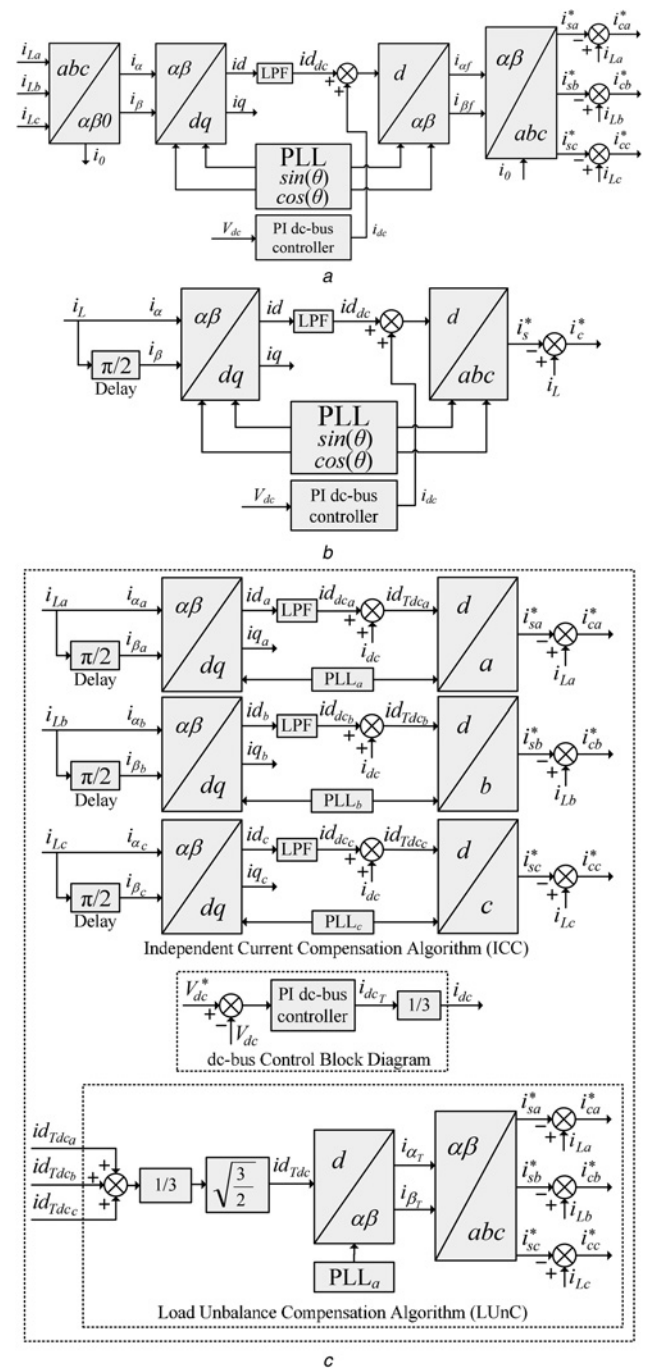


Fig. 2 SRF-based algorithms

a Three-phase conventional algorithm

b Single-phase algorithm

c SRF algorithms for independent control (ICC) and for load unbalance compensation (LUnC)

The modified SRF-based algorithm applied to single-phase systems is shown in Fig. 2b. Thereby, measuring the single-phase load current i_L , it is possible to provide the two-phase stationary reference frame $\alpha\beta$ quantities (i_α , i_β) as given by (3). Thus, the measured load current is treated as the α -axis coordinate ($i_\alpha = i_L$) of the fictitious two-phase stationary reference frame ($\alpha\beta$ -axes). Subsequently, i_α is phase delayed by $\pi/2$ radians, producing the fictitious β -axis coordinate (i_β). Therefore a new two-phase system can be studied in $\alpha\beta$ -axes and the d quantity (id) of the SRF is obtained by (4). Using an LPF, the dc component (id_{dc}) of id -axis can be obtained. The current i_{dc} (Fig. 2) represents

the output of the dc-bus proportional-integral (PI) controller, where the gains P and I are defined as K_{Pdc} and K_{Idc} , respectively. This controller is responsible to control the dc-bus voltage, providing the compensation of the losses related to the filtering inductances and switching devices. In other words, the PI output signal i_{dc} represents the total active current required to maintain the dc-bus voltage.

The fundamental current reference i_s^* is achieved directly from the synchronous rotating dq -axes as given by (5), where θ is the utility phase angle. Finally, the compensation current reference of the SAPF is given by (6), which includes the compensation of the reactive and harmonic components of the load current

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \begin{bmatrix} i_L(\theta) \\ i_L(\theta - \pi/2) \end{bmatrix} \quad (3)$$

$$id = i_\alpha \cos \theta + i_\beta \sin \theta \quad (4)$$

$$i_s^* = (id_{dc} + i_{dc}) \cos \theta \quad (5)$$

$$i_c^* = i_L - i_s^* \quad (6)$$

3.2 SRF-based algorithms applied to the SAPF

Once the phase currents of the three-phase SAPF discussed in this work can be independently controlled, the algorithm shown in Fig. 2b is used for composing the two SRF algorithms shown in Fig. 2c, which are used to implement two compensation filtering strategies.

The algorithms could provide independent control compensation (ICC), as well as LUnC. For the ICC algorithm, the compensation current references (i_{ca}^* , i_{cb}^* , i_{cc}^*) can be obtained by using (6). In this case, the total current i_{dc} obtained from the dc-bus controller (Fig. 2c) is divided by 3 and the result (i_{dc}) is added to the respective currents ($id_{dc,a,b,c}$). For the LUnC algorithm, the new current references are obtained by generating the source current references given by (7), the algorithm of which is also shown in Fig. 2c, where $i_{\alpha T}$, $i_{\beta T}$ and id_{Tdc} are given by (8), (9) and (10), respectively

$$\begin{bmatrix} i_{sa}^* \\ i_{sb}^* \\ i_{sc}^* \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & 0 \\ -1/2 & \sqrt{3}/2 \\ -1/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} i_{\alpha T} \\ i_{\beta T} \end{bmatrix} \quad (7)$$

$$i_{\alpha T} = id_{Tdc} \cos(\theta) \quad (8)$$

$$i_{\beta T} = id_{Tdc} \sin(\theta) \quad (9)$$

$$id_{Tdc} = \sqrt{\frac{3}{2}} \left(\frac{id_{Tdc,a} + id_{Tdc,b} + id_{Tdc,c}}{3} \right) \quad (10)$$

Finally, the three-phase APF compensation current references are thus found as

$$i_{ca}^* = i_{La} - i_{sa}^*; \quad i_{cb}^* = i_{Lb} - i_{sb}^*; \quad i_{cc}^* = i_{Lc} - i_{sc}^* \quad (11)$$

3.3 State-feedback current controller

Since the compensation current references are extracted from SRF algorithms, they must be synthesised by the SAPF. Thus, the mathematical model of the SAPF needs to be obtained in

order to achieve an adequate design procedure used to calculate the PI gains of the current controllers.

The dynamic model of the SAPF is obtained from a single-phase equivalent circuit shown in Fig. 3a, in which the secondary transformer impedances are referred to the primary side, where L_{L_1} is the transformer leakage inductance, L_f is the filter inductance; R_{L_1} and R_{L_f} are their respective resistances; n is the transformer ratio ($n = 1$) and v_s is the phase voltage. Fig. 3b presents the block diagram of the current controller model used for each one of the three full-bridge APF topologies, and the per-phase closed-loop transfer function of the SAPF is represented by (12), where K_{Pp} and K_{Ip} are the proportional and the integral gains of the current PI controller; $L_{eq} = L_f + n^2 L_{L_1}$ is the equivalent inductance; $R_{eq} = R_{L_f} + n^2 R_{L_1}$ is the equivalent resistance; K_{PWM} is the PWM gain [39] and V_{dc} is the dc-bus voltage

$$\frac{i_{c,a,b,c}(s)}{i_{c,a,b,c}^*(s)} = \frac{K_{PWM} V_{dc} (K_{Pp}s + K_{Ip})}{L_{eq}s^2 + (K_{Pp} K_{PWM} V_{dc} + R_{eq})s + K_{Ip} K_{PWM} V_{dc}} \quad (12)$$

3.4 Loop modelling of the dc-bus (single-phase and three-phase systems)

In this sub-section the dc-bus control loop modelling is developed for single-phase systems. After that, it is extended to three-phase systems. The dc-bus voltage control loop is used to maintain the dc-bus voltage at a constant value. Thus, controlling the active current drained from the utility grid by the SAPF, the switching losses of the PWM converter, as well as the losses related to the filtering elements can be compensated.

The instantaneous active power of a single-phase system is defined by

$$p_{in} = v_s \cdot i_s = V_p \sin(\theta) \cdot I_p \sin(\theta) \quad (13)$$

where v_s and i_s represent, respectively, the single-phase voltage and current of the utility grid, V_p and I_p are their respective amplitudes and θ is the phase-angle of the grid.

It is assumed that the electric grid voltage v_s is sinusoidal and the SAPF compensates all the harmonic components of the load current. As a result, the compensated source current i_s can also be considered sinusoidal.

This single-phase system can be represented by a fictitious sinusoidal and balanced three-phase system, which can be described by voltage and current magnitudes into the two-phase orthogonal stationary frame ($\alpha\beta$ -axes). Thus, the resultant fictitious instantaneous power is given by

$$p'_{\alpha\beta} = v'_\alpha \cdot i'_\alpha + v'_\beta \cdot i'_\beta = V_p \sin(\theta) \cdot I_p \sin(\theta) + V_p \sin(\theta - \pi/2) \cdot I_p \sin(\theta - \pi/2) \quad (14)$$

The parameters used for computing $p'_{\alpha\beta}$ were extracted from the single-phase measurements. Therefore the real average input power is equal to the average power obtained from $p'_{\alpha\beta} \cdot (\bar{p}'_{\alpha\beta})$ divided by two, that is

$$\bar{p}'_{in} = \frac{\bar{p}'_{\alpha\beta}}{2} \quad (15)$$

The same reasoning can be expanded if the single phase is

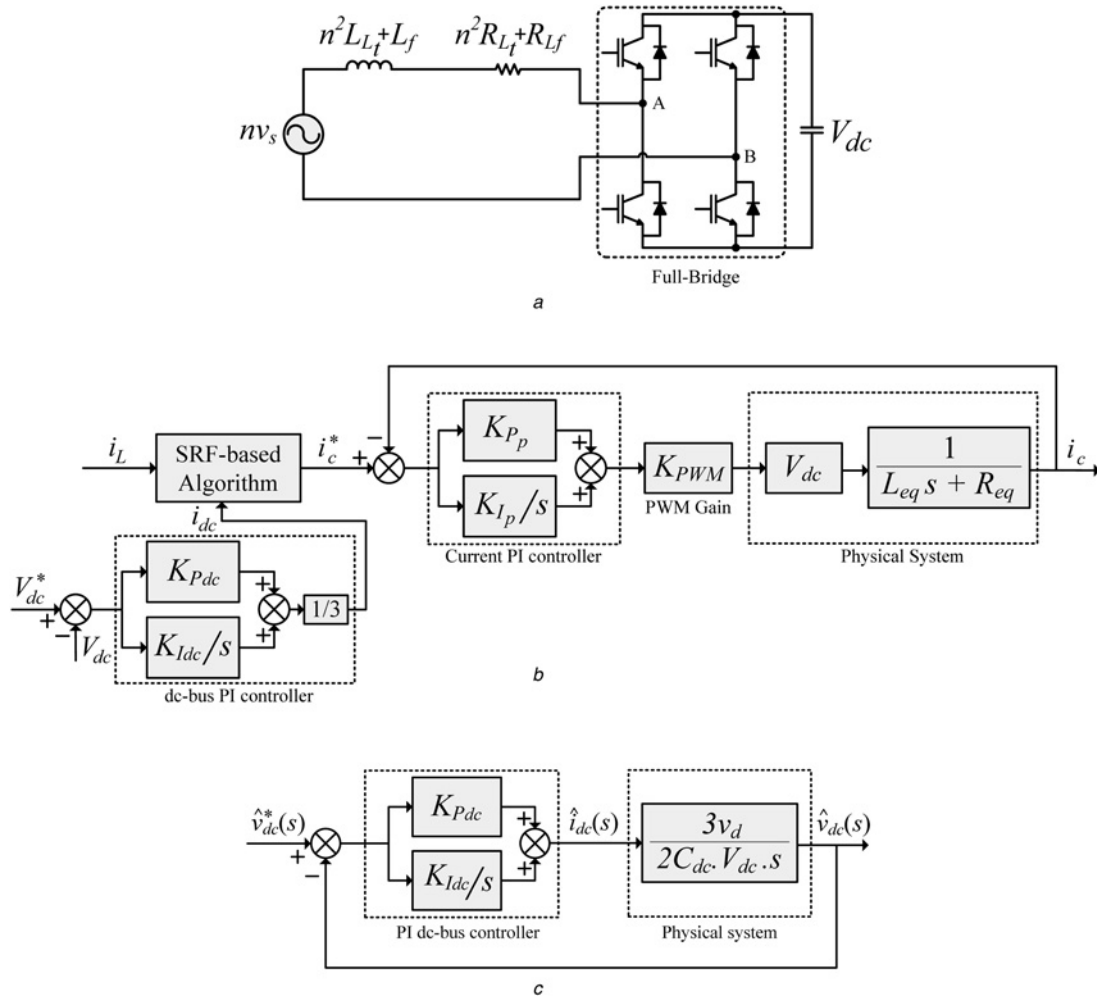


Fig. 3 Dynamic model of the SAPF, block diagram of the current controller model and dc-bus voltage control loop

a Single-phase equivalent circuit of the SAPF
 b Block diagram of the SAPF current controller
 c Block diagram of the dc-bus voltage control loop

represented by a fictitious three-phase system in the SRF (dq -axes). Assuming that, both the quantities of voltage and current are balanced and harmonic free, they can be represented in the dq -axes only by continuous components. In this case, the real average input power in the dq -axes is obtained by

$$\bar{p}_{in} = \frac{v_d \cdot i_d}{2} \quad (16)$$

By this means, v_d and i_d represent the voltage and current into the fictitious SRF, represented, respectively, by (17) and (18)

$$v_d = v'_\alpha \cdot \cos(\theta_{pll}) + v'_\beta \cdot \sin(\theta_{pll}) \quad (17)$$

$$i_d = i'_\alpha \cdot \cos(\theta_{pll}) + i'_\beta \cdot \sin(\theta_{pll}) \quad (18)$$

where θ_{pll} is the estimated phase angle of the utility grid voltage ($\theta_{pll} = \theta$) obtained using a PLL scheme.

Similarly, the power supplied to the load is given by

$$p_L = \frac{v_d \cdot i_{dL}}{2} = \frac{v_d \cdot (i_{dL_{dc}} + i_{dL_{ac}})}{2} \quad (19)$$

where i_{dL} represents the load current in the direct

axis; $i_{dL_{dc}}$ and $i_{dL_{ac}}$ represent the load average and alternating currents in the direct axis of the SRF.

Considering that $i_{d_{dc}} = i_{dL_{dc}}$, the instantaneous active power delivered to the load can be written as

$$p_L = \bar{p}_{in} + \frac{v_d \cdot i_{dL_{ac}}}{2} \quad (20)$$

Subtracting (20) from (16), the power flowing through the SAPF is obtained as

$$p_{APF} = p_L - \bar{p}_{in} = \frac{v_d \cdot i_{dL_{ac}}}{2} = p_{ac} \quad (21)$$

If there is an active portion of the power drawn by the SAPF, which is necessary to compensate for losses in the switching and filtering elements, (21) can be rewritten as

$$p_{APF} = \bar{p}_{in_{apf}} + p_{ac} = \frac{v_d \cdot i_{dc_{apf}}}{2} + p_{ac} \quad (22)$$

where $i_{dc_{apf}}$ represents the active current flowing through the SAPF.

Thus, considering the dc-bus PI controller (Fig. 3c), the open loop and the closed loop transfer functions are represented, respectively, by (29) and (30)

$$G_{OL}(s) = \frac{3(v_d K_{Pdc} s + v_d K_{Idc})}{2C_{dc} V_{dc} s^2} \quad (29)$$

$$G_{CL}(s) = \frac{\hat{v}_{dc}(s)}{\hat{v}_{dc}^*(s)} = \frac{3v_d K_{Pdc} s + 3v_d K_{Idc}}{2C_{dc} V_{dc} s^2 + 3v_d K_{Pdc} s + 3v_d K_{Idc}} \quad (30)$$

4 PLL system

The coordinates of unit vector used in the SRF algorithms ($\sin \theta$ and $\cos \theta$) could be obtained from a three-phase PLL system. However, it was chosen the implementation of three single-phase PLL systems, taking into account that each one of the full-bridge inverter can be considered as a single-phase APF unit, sharing the same dc-bus voltage. Thus, if one of the three single-phase APF units is damaged the other two units could operate normally. The single PLL system used in this work has been described in detail in [34] and its structure is shown in Fig. 4a.

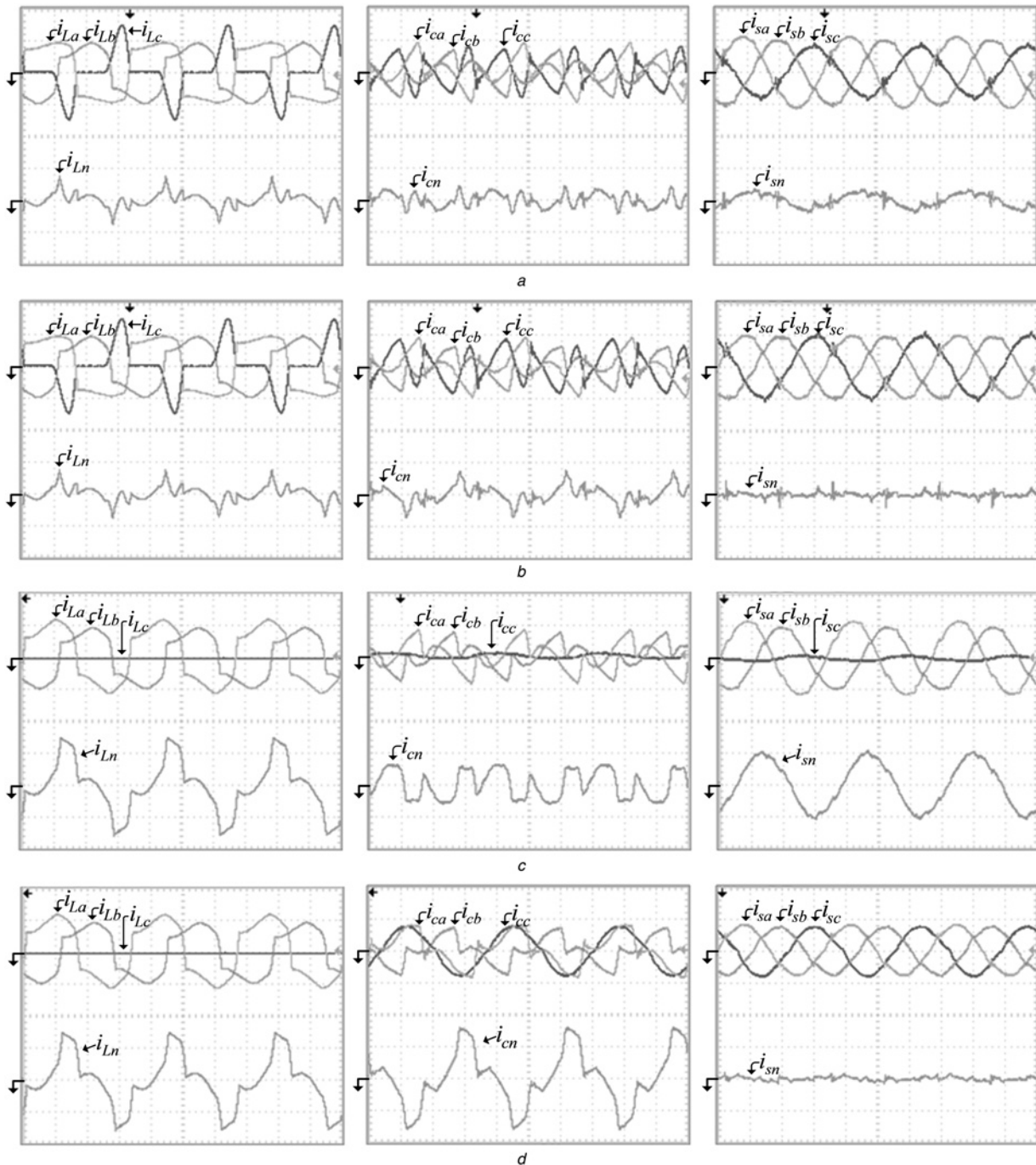


Fig. 5 Single-phase loads 2 and 3 (20 A/div, 5 ms/div)

- a Load currents ($i_{La,b,c}$), compensation currents ($i_{ca,b,c}$), source currents ($i_{sa,b,c}$) and neutral currents (i_{Ln} , i_{cn} , i_{sn}), for ICC (load 2)
- b Load currents, compensation currents, source currents and neutral currents, for LUnC (load 2)
- c Load currents, compensation currents, source currents and neutral currents, for ICC (load 3)
- d Load currents, compensation currents, source currents and neutral currents, for LUnC (load 3)

4.1 Self-tuning filter (STF)-pPLL algorithm

The pPLL system shown in Fig. 4a is not robust enough and could be affected by disturbances of the utility voltage, such as voltage harmonics. These disturbances interfere on the performance of the PLL and, consequently, the quality of PLL output signals used in the SAPF is affected.

To overcome this problem, the input voltage can be filtered by means of the STF shown in Fig. 4b, which is implemented and described in detail in [40]. In the STF scheme, the cut-off frequency ω_c represents the nominal utility angular frequency. The STF was initially proposed for application in three-phase systems. In this work, the STF operates in conjunction with the pPLL system as shown in Fig. 4b, resulting in the STF-pPLL algorithm. In other words, the STF is employed to extract the fundamental components of the measured utility phase voltages, and the pPLL is used for obtaining the correct phase angle of the utility. In addition, the reference angular frequency of the pPLL (ω^*) is used to adapt the cut-off frequency of the STF, considering that variations in the mains frequency could result in phase-angle errors. Thus, as can be seen from Fig. 4b, the input signals of the STF comprise two quadrature voltages (v_{α}, v_{β}) and the output voltages ($v_{\alpha f}, v_{\beta f}$) represent the respective filtered STF quadrature signals, which are used

as inputs of the pPLL shown in Fig. 4c. The parameter K shown in Fig. 4b defines the selectivity of the filter, that is, the smaller the value of K , the more selective the filter.

5 Experimental results

The SAPF was implemented and experimentally tested according to the topology shown in Fig. 1. The performance of the filter is available through the experiments based on the digital signal processor (DSP) TMS320F28335 from Texas Instruments, which was used to implement the current SRF-based algorithms (Fig. 2) and the STF-pPLL algorithm (Fig. 4). The switching frequencies of the 3F-B converters were set at 20 kHz, and all the parameters used in the SAPF tests are shown in Table 1. Table 2 presents the loads used in the experiments. Each full-bridge inverter employs four IGBT modules (SK40GB 123 Semikron) with their proper drivers. Six current sensors, model LEM LA 100-P, are used to measure both the load currents and the SAPF compensation currents. The input ac voltages and the dc-bus voltage are measured using four voltage sensors model LEM LV 25-P. The approach used for tuning the PI controllers, such as the current controller and the dc-bus controller gains, is based

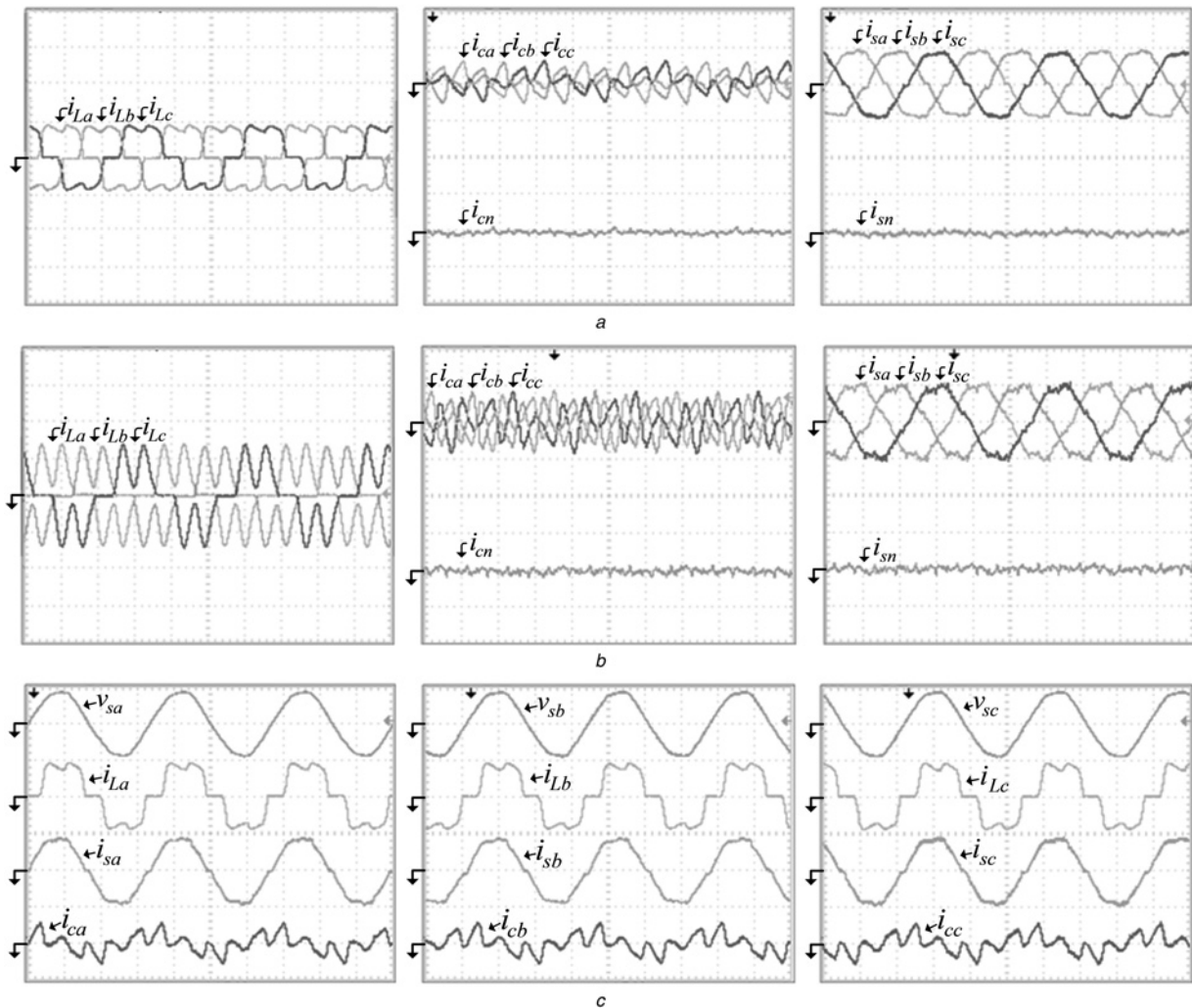


Fig. 6 Three-phase loads (20 A/div, 200 V/div, 5 ms/div)

a Load 1: load currents (i_{La}, i_{Lb}, i_{Lc}), compensation currents (i_{ca}, i_{cb}, i_{cc}), source currents (i_{sa}, i_{sb}, i_{sc}) and neutral currents (i_{Ln}, i_{cn}, i_{sn})

b Load 2: load currents, compensation currents, source currents and neutral currents

c Load 1: input voltages v_{sa}, v_{sb}, v_{sc} and currents $i_{La}, i_{Lb}, i_{Lc}, i_{sa}, i_{sb}, i_{sc}, i_{ca}, i_{cb}, i_{cc}$

Table 3 THD of the load and the source currents

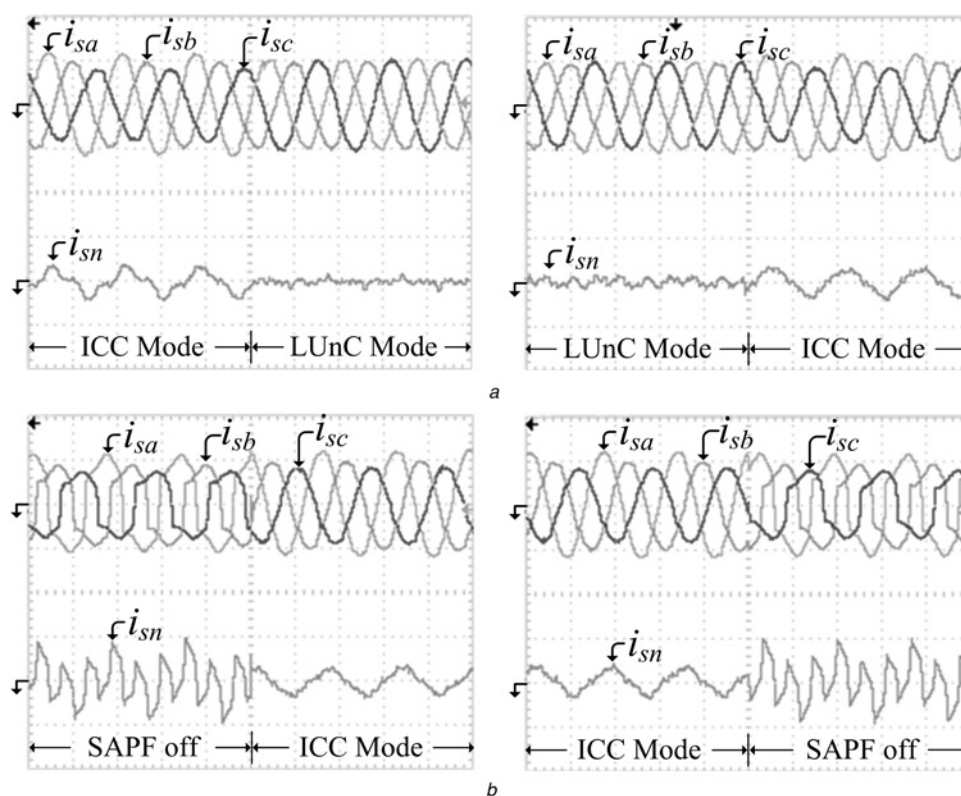
Single and three-phase loads		Current total harmonic distortion (THD %)					
		i_{La}	i_{Lb}	i_{Lc}	i_{sa}	i_{sb}	i_{sc}
single-phase load 1	ICC	25	25.6	25.8	3.66	3.83	3.70
	LUnC				4.60	3.63	2.49
single-phase load 2	ICC	34.6	23.8	77.2	4.13	4.07	9.04
	LUnC				5.69	4.23	8.30
single-phase load 3	ICC	25	25.6	—	3.65	3.73	—
	LUnC				5.06	4.28	4.79
three-phase load 1	LUnC	25	25	25.8	6.17	6.54	6.09
three-phase load 2	LUnC	55.2	55	57.6	8.71	9.13	8.76

on the frequency response method via Bode diagrams taking into account the phase margin and the 0 dB gain crossover frequency as design specifications. Table 1 presents the tuning parameters of the controllers. For the PI controller, the trapezoidal method was considered in the controller discretisation. The d -axis components of the SRF-based algorithms are filtered using LPFs. First-order filters with cut-off frequency equal to 1 Hz was used. The parameter K_{PWM} (Table 1) represents the static gain of the PWM converters. It is defined taking into account the peak value of the PWM triangular carrier [39].

The experimental results for the SAPF operating both for independent current compensation and for LunC strategies are shown in Fig. 5, considering the single-phase loads 2 and 3 presented in Table 2. Figs. 5a and c show, for ICC operation mode, the load currents (i_{La} , i_{Lb} , i_{Lc}), the parallel compensation currents (i_{ca} , i_{cb} , i_{cc}), the compensated source currents (i_{sa} , i_{sb} , i_{sc}) and their respective neutral currents (i_{Ln} , i_{cn} , i_{sn}). As can be noted, the source fundamental

neutral current (i_{sn}) flows through the neutral conductor, whereas the zero sequence harmonic components flow through the neutral conductor of the SAPF ($i_{cn} = i_{Ln}$). Figs. 5b and d show, for LUnC operation mode, the load currents, the parallel compensation currents, the compensated source currents and the respective neutral currents. As can be observed, the source fundamental neutral current is approximately equal to zero ($i_{sn} \cong 0$), whereas the source currents (i_{sa} , i_{sb} , i_{sc}) are balanced. Thus, both zero and negative sequence components of current were totally compensated and all fundamental and harmonic components flowed through the SAPF neutral conductor ($i_{cn} = i_{Ln}$).

The experimental results for the SAPF performing the current compensation considering the three-phase loads 1 and 2 presented in Table 2 are shown in Fig. 6. The load currents, the compensation currents, the source currents and their respective neutral currents ($i_{cn} = i_{sn} \cong 0$) are shown in Figs. 6a and b, respectively. Fig. 6c shows, separately per

**Fig. 7** SAPF transition modes (single-phase load 1 (20 A/div, 10 ms/div))

a Source currents ($i_{sa,b,c}$) and neutral current (i_{sn}) for ICC to LUnC and LUnC to ICC

b Source currents ($i_{sa,b,c}$) and neutral current (i_{sn}) for SAPF off to ICC mode and ICC mode to SAPF off

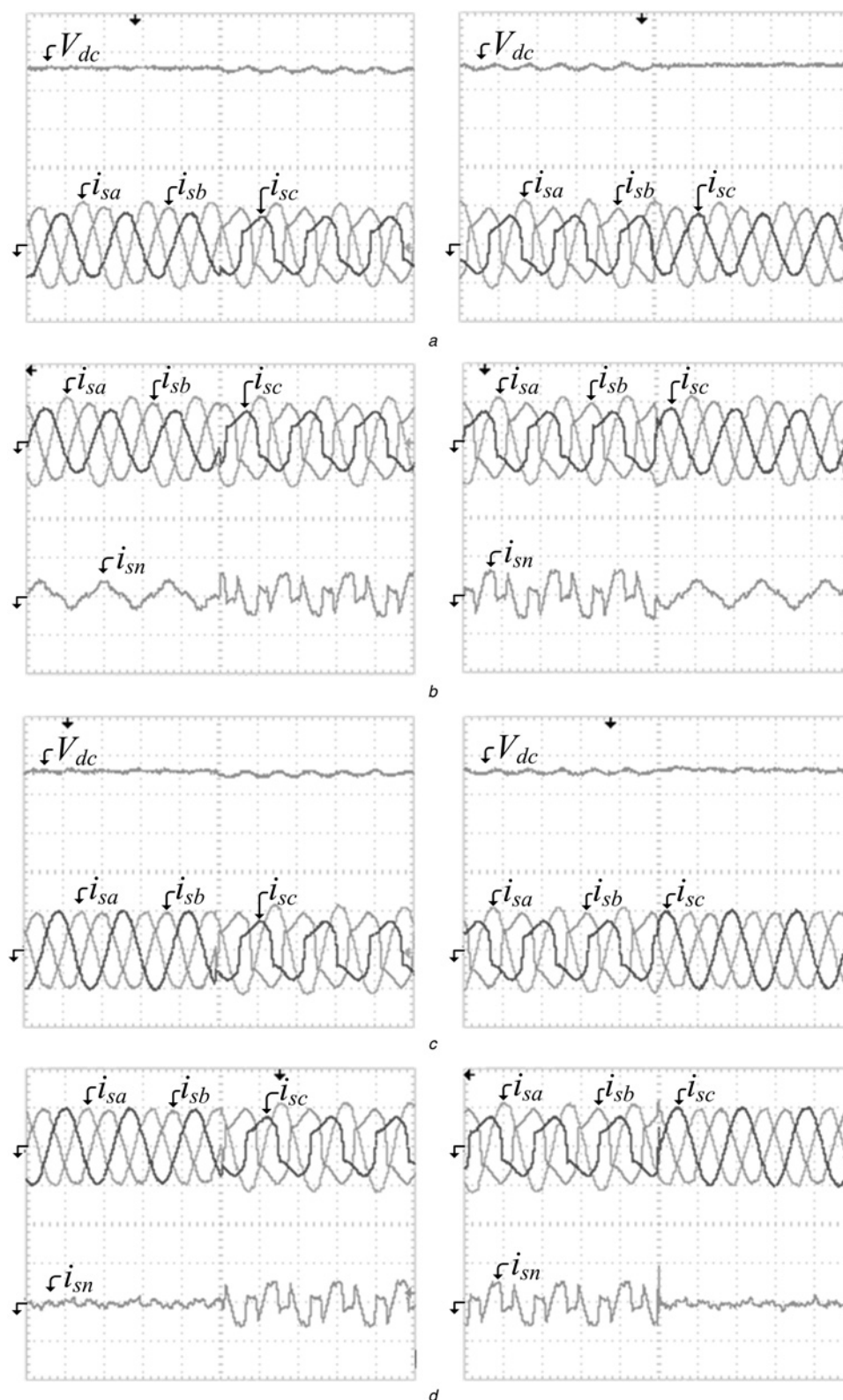


Fig. 8 SAPF transition modes (for single-phase load 1 (20 A/div and 50 V/div, 10 ms/div))

- a Source currents ($i_{sa,b,c}$) and dc-bus voltage (V_{dc}) for normal ICC operation to two inverters fail to operate
b Source currents ($i_{sa,b,c}$) and neutral current (i_{sn}) for normal ICC operation to two inverters fail to operate
c Source currents ($i_{sa,b,c}$) and dc-bus voltage (V_{dc}) for normal LUnC operation to two inverters fail to operate
d Source currents ($i_{sa,b,c}$) and neutral current (i_{sn}) for normal LUnC operation to two inverters fail to operate

phase, the input voltages $v_{sa,b,c}$, and the currents $i_{La,b,c}$, $i_{sa,b,c}$ and $i_{ca,b,c}$, for the three-phase load 1.

The total harmonic distortion (THD) of the load and the source currents are presented in Table 3, considering both

the SAPF operation modes (ICC and LUnC). As can be noted, the THD of the source currents was strongly reduced for all types of loads used in the tests. Table 3 also shows that the THD of the source currents with the three-phase

load (load 1) is higher than with single-phase load (load 1), considering the LUnC strategy. This is justified because of the fact that the high current derivative of the SAPF reference currents does not happen at the same time for the two mentioned types of loads. Thereby, the effect related to high current derivative is different for the two loads. Thus, for the three-phase load, the PI current controller suffers with higher current derivatives than the single-phase loads and, therefore, the THD of the source currents is increased.

The flexibility of the SAPF operation can be observed in Fig. 7a, when the change of the operation modes from ICC to LUnC and from LUnC to ICC occurs. Fig. 7b shows the additional results from non-compensating mode to compensating mode considering the SAPF operating the ICC mode. Fig. 8 shows the SAPF operations, considering that two inverters fail to operate at the same time. In Figs. 8a and b ICC strategy is adopted. As can be observed, up to the fails occurrence of the inverters connected to the phase 'b' and 'c', the source currents $i_{sa,b,c}$ are sinusoidal although unbalanced. Thereby, only the current related to the phase 'a' remains compensated. In Figs. 8c and d LUnC strategy is adopted. As can be observed, up to the fails occurrence the source currents $i_{sa,b,c}$ are sinusoidal and balanced. After that, only phase 'a' remains normally operating by means of the ICC strategy. Thus, only the current related to phase 'a' remains compensated.

6 Conclusions

This paper proposed the SRF-based algorithms for obtaining the compensation current references for SAPF applied to three-phase four-wire systems, allowing current harmonic suppression, load reactive power compensation, power factor improvement and LUnC, as well. The SRF-based algorithms were evaluated under balanced and unbalanced load conditions using an SAPF constructed by means of three full-bridge inverters. Thus, it was possible to control each phase of the three-phase four-wire system independently, so that the compensated source currents became sinusoidal, although unbalanced for load unbalance condition. This means that only the harmonics and reactive power of the load were compensated and both zero and negative fundamental sequence components of current were not taken into account. On the other hand, the compensation of the load unbalances enabled the source currents to become sinusoidal and balanced. As can be noted, the controllability degree of the SAPF was increased ensuring more accuracy in the current unbalance compensation.

In addition, once the SAPF uses three independent active filtering units, it can represent an important advantage. For instance, if one or two inverters fail to operate, the SAPF can remain operating normally.

It was also demonstrated that the experimentally obtained results agree with good approximation with the theoretically predicted results.

7 References

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